Lecture 10
The BFV Scheme

SEC 595: Encrypted Computing

Roadmap

Lecture 8 defined homomorphic encryption and Lecture 7 built an LWE scheme that only encrypts. This lecture assembles the first lattice scheme that adds and multiplies: BFV (Fan and Vercauteren, 2012). Every ingredient already exists — the ring R_q (Lecture 3), the scaling factor Δ (Lecture 1), gadget decomposition (Lecture 4), and RLWE (Lecture 6). Sections 1–4 give the spaces, encoding, keys, encryption and the correctness condition. Section 5 shows that addition is cheap. Sections 6–7 treat multiplication, its noise growth, and relinearization. Section 8 runs everything on a small instance, and Section 9 places BFV beside BGV and CKKS. Sections 1–5 form the first session and Sections 6–9 the second.

1. Spaces and Parameters

1.1 The Interface

Primitive	**Returns**
ParamGen(λ)	(n, t, q) and the distributions
KeyGen	SK (kept), PK and EK (published)
Encrypt(PK, M)	$\mathrm{ct} \in R_q \times R_q$
Decrypt(SK, ct)	$M \in R_t$
EvalAdd($\mathrm{ct}^{(1)}, \mathrm{ct}^{(2)}$)	an encryption of $M^{(1)} + M^{(2)}$
EvalMult($\mathrm{ct}^{(1)}, \mathrm{ct}^{(2)}$)	an encryption of $M^{(1)}M^{(2)}$
Relinearize(EK, ct^*)	a two-element ct encrypting the same M

BFV is **levelled** (Lecture 8): it evaluates circuits up to a multiplicative depth fixed when (n, t, q) is chosen. Relinearize computes nothing; it repairs the shape of the output of EvalMult.

1.2 Two Rings

Definition 1 (Plaintext and ciphertext spaces)**.** Let n be a power of two and $q \gg t \geq 2$. Plaintexts lie in $R_t = \mathbb{Z}_t[X]/(X^n + 1)$ and ciphertexts in $R_q \times R_q$, where $R_q = \mathbb{Z}_q[X]/(X^n + 1)$.

Coefficients are taken in the centered range, and $[\,\cdot\,]_q$ denotes reduction into it (Lecture 1). For $v \in R_q$, $\|v\|$ is the largest absolute coefficient, and $\|ab\| \leq n\|a\|\|b\|$ (Lecture 3); every noise bound below is an application of this inequality. Since $|R_t| = t^n$ and $|R_q \times R_q| = q^{2n}$, at $n = 4$, $t = 4$, $q = 2^{14}$ there are $4^4 = 256$ plaintexts against 2^{112} ciphertexts: encryption has ample room to be randomised.

Example 1 (Plaintexts at $(n,t)=(4,5)$)**.** For $t=5$ the centered range is $\{-2,-1,0,1,2\}$, and a plaintext is a polynomial of degree at most 3 with coefficients in it. Hence

$$1+2X+X^2-X^3, \qquad -1-2X-X^2+2X^3, \qquad 1+X+X^2+X^3$$

all lie in R_5, and $|R_5|=5^4=625$.

1.3 Distributions and Parameter Choice

Symbol	Distribution	Used for
R_2	ternary, coefficients uniform in $\{-1,0,1\}$	SK and u
$\mathcal{X}$	discrete Gaussian (μ,σ,β)	e, e_1, e_2
R_q	uniform over R_q	the mask a

The standard choice is $\mu=0$, $\sigma=8/\sqrt{2\pi}\approx 3.2$ and tail-cut $\beta=\lfloor 6\sigma\rceil=19$ (Lecture 6, §1.3). Because of the tail-cut, $\|e\|\le\beta$ is treated as a hard bound, and noise analysis becomes arithmetic.

The security parameter λ is a work parameter: $\lambda=128$ means the best known attack costs about 2^{128} operations. The rule of thumb is to minimise n, t and q subject to the computation fitting. A larger n costs time and size and buys security; a larger q buys depth but costs security at fixed n; a larger t buys plaintext room but shrinks the budget $q/2t$ and inflates the growth factor $2tn^2$ of Section 6.

2. Encoding

Encoding maps data into R_t. It uses no keys and no randomness.

Naive encoding puts m in the constant term, $M=m+0\cdot X+\cdots+0\cdot X^{n-1}$, and decodes by reading it back. It wastes $n-1$ coefficients, and the single coefficient must hold every intermediate value, so t must exceed the largest one — and a large t costs noise budget.

Definition 2 (Base-k encoding)**.** Write $m=(a_{n-1}\cdots a_1a_0)_k$ and set $M=a_{n-1}X^{n-1}+\cdots+a_1X+a_0$. Decode by evaluating, $m=M(k)$, reading coefficients as non-negative representatives in $\{0,\dots,t-1\}$.

Coefficients now start at most $k-1$ instead of m. But multiplication raises degrees, and degree wraps at n.

Proposition 1 (Compatibility)**.** *Base-k encoding is compatible with a circuit when, at every intermediate result, (a) no coefficient reaches t in absolute value, and (b) the degree stays below n.*

If (a) fails, a coefficient is corrupted modulo t. If (b) fails, $X^n\equiv -1$ folds high terms onto low ones with a sign flip. Boolean circuits over $\mathbb{Z}_2$ tolerate (a) by design, since reduction modulo 2 is the intended semantics.

Example 2 (Encoding by hand, $n=4$)**.** *Base* 2. $13=1101_2$, so $M=1+X^2+X^3$, and $M(2)=1+4+8=13$.

Base 3. $13=111_3$, so $M=1+X+X^2$, and $M(3)=1+3+9=13$.

Base 2 has coefficients at most 1, so it leaves more room before (a) fails; base 3 uses degree 2 instead of 3, so it leaves more room before (b) fails.

$m = 21$ *in base* 2. $21 = 10101_2$ needs X^4. In R_t, $1 + X^2 + X^4 = 1 + X^2 - 1 = X^2$, which decodes to 4, not 21.

3. Key Generation

Definition 3 (KeyGen)**.** Sample $\mathrm{SK} \leftarrow R_2$, $a \leftarrow R_q$ and $e \leftarrow \mathcal{X}$, and set

$$\mathrm{PK}_0 = \big[-(a \cdot \mathrm{SK} + e)\big]_q, \qquad \mathrm{PK}_1 = a.$$

The key identity is one line:

$$\mathrm{PK}_0 + \mathrm{PK}_1 \cdot \mathrm{SK} \;=\; -(a\mathrm{SK} + e) + a\mathrm{SK} \;=\; -e. \tag{1}$$

The public key, evaluated at the secret, is small. The pair $(\mathrm{PK}_0, \mathrm{PK}_1)$ is one RLWE sample, so distinguishing it from uniform is decision RLWE (Lecture 6, §2.2). The evaluation key EK is also produced here; it is defined in Section 7.

4. Encryption and Decryption

4.1 The Scaling Factor

Let $\Delta = \lfloor q/t \rfloor$ and write $q = \Delta t + [q]_t$ with $0 \le [q]_t < t$. Multiplying by Δ places the message in the top $\log_2 t$ bits and leaves the bottom $\log_2(q/t)$ bits for noise (Lecture 1). Decryption scales by t/q and *rounds*. The residue $[q]_t$ vanishes exactly when $t \mid q$.

4.2 The Algorithms

Definition 4 (Encrypt and Decrypt)**.** Encrypt(PK, M): sample $u \leftarrow R_2$ and $e_1, e_2 \leftarrow \mathcal{X}$, and return $\mathrm{ct} = (c_0, c_1)$ with

$$c_0 = \big[\mathrm{PK}_0 \cdot u + e_1 + \Delta M\big]_q, \qquad c_1 = \big[\mathrm{PK}_1 \cdot u + e_2\big]_q. \tag{2}$$

Decrypt$(\mathrm{SK}, \mathrm{ct})$: return

$$M = \left[\left\lfloor \frac{t\,\big[c_0 + c_1 \cdot \mathrm{SK}\big]_q}{q} \right\rceil \right]_t. \tag{3}$$

The three small polynomials u, e_1, e_2 are drawn per encryption and never needed again. They make encryption randomised, as IND-CPA requires (Lecture 8, §1.5).

4.3 The Decryption Identity

Proposition 2 (Decryption identity)**.** *For a fresh ciphertext,*

$$c_0 + c_1 \cdot \mathrm{SK} \;=\; \Delta M + v \pmod q, \qquad v \;=\; -e \cdot u + e_1 + e_2 \cdot \mathrm{SK}.$$

Proof. Substitute (2) and then the definition of PK:

$$\begin{aligned} c_0 + c_1\mathrm{SK} &= -(a\mathrm{SK} + e)u + e_1 + \Delta M + (au + e_2)\mathrm{SK} \\ &= -au\mathrm{SK} - eu + e_1 + \Delta M + au\mathrm{SK} + e_2\mathrm{SK} \;=\; \Delta M - eu + e_1 + e_2\mathrm{SK}. \quad \square \end{aligned}$$

The mask $au\mathrm{SK}$ cancels. The encryptor creates it inside $\mathrm{PK}_0 u$ without knowing SK; the decryptor creates it inside $c_1\mathrm{SK}$ without knowing u. What remains is the scaled message plus the **noise** v.

Proposition 3 (Fresh noise)**.** *If* $\|e\|, \|e_1\|, \|e_2\| \le \beta$ *and* $\|u\|, \|\mathrm{SK}\| \le 1$*, then* $\|v\| \le n\beta + \beta + n\beta = 2n\beta + \beta$*. If* u *and* SK *are only* β*-bounded,* $\|v\| \le 2n\beta^2 + \beta$*.*

Proof. Apply $\|ab\| \le n\|a\|\|b\|$ to eu and $e_2\mathrm{SK}$, and add $\|e_1\| \le \beta$. □

At $n = 4096$, $\beta = 19$:

$$2n\beta + \beta = 155\,667 \approx 2^{17.25}, \qquad 2n\beta^2 + \beta = 2\,957\,331 \approx 2^{21.50}.$$

The second, conservative bound is the one quoted in the literature, because it also covers variants with $u \leftarrow \mathcal{X}$. At $\beta = 1$ the two coincide.

4.4 The Correctness Condition

Proposition 4 (Correctness)**.** *If* $c_0 + c_1\mathrm{SK} = \Delta M + v \pmod q$ *with* $M \in R_t$ *and*

$$\|v\| \;<\; \frac{q}{2t},$$

then Decrypt *returns* M*.*

Proof. Over $\mathbb{Z}$, $c_0 + c_1\mathrm{SK} = \Delta M + v + qr$ for some integer polynomial r. Scaling by t/q gives $\frac{t\Delta}{q}M + \frac{t}{q}v + tr$, and $\frac{t\Delta}{q} = 1 - \frac{[q]_t}{q}$ differs from 1 by a negligible amount. Rounding removes $\frac{t}{q}v$ when each coefficient is below $\frac{1}{2}$, which is the hypothesis; reduction modulo t removes tr. □

Only the rounding can fail, and the failure is *silent*: a coefficient comes back wrong, with no warning. The gap between $\|v\|$ and $q/2t$ is the **noise budget**; every homomorphic operation spends part of it.

Example 3 (One failure)**.** Take $t = 8$, $q = 1024$, one coefficient $m = 5$. Then $\Delta = \lfloor 1024/8 \rfloor = 128$, $\Delta m = 640$, and the bound is $q/(2t) = 1024/16 = 64$.

$$\begin{aligned} v = 30: &\quad \tfrac{8}{1024}(640 + 30) = \tfrac{5360}{1024} = 5.234375 \;\longrightarrow\; 5 \quad \checkmark \\ v = 70: &\quad \tfrac{8}{1024}(640 + 70) = \tfrac{5680}{1024} = 5.546875 \;\longrightarrow\; 6 \quad \times \end{aligned}$$

At $v = 70 > 64$ the coefficient comes back as 6. The output is a valid element of $\mathbb{Z}_8$, so the decryptor cannot tell.

5. EvalAdd

Definition 5 (EvalAdd)**.**

$$\mathrm{EvalAdd}(\mathrm{ct}^{(1)}, \mathrm{ct}^{(2)}) = \Big(\big[c_0^{(1)} + c_0^{(2)}\big]_q,\ \big[c_1^{(1)} + c_1^{(2)}\big]_q \Big).$$

The result has the shape of a fresh encryption of $M^{(1)} + M^{(2)}$ with randomness $u^{(1)} + u^{(2)}$ and $e_i^{(1)} + e_i^{(2)}$, so it can be fed back in.

Proposition 5 (Additive noise growth)**.** $c_0^{(3)} + c_1^{(3)}\mathrm{SK} = \Delta\big(M^{(1)} + M^{(2)}\big) + (v_1 + v_2)$*, so* $\|v_{\mathrm{add}}\| \le \|v_1\| + \|v_2\|$.

Proof. Add the two decryption identities. □

Thus L fresh ciphertexts sum to noise at most $L(2n\beta+\beta)$. At $n = 4096$, $\beta = 19$, fresh noise is about $2^{17.25}$, so a budget of 2^{178} permits about 2^{160} additions. To add a plaintext M', add $\Delta M'$ to c_0 and leave c_1; the noise is unchanged. Addition is essentially free: the whole difficulty is multiplication.

6. EvalMult

6.1 Multiplying the Identities

Read ct as $c_0 + c_1 S \in R_q[S]$, so that decryption evaluates at $S = \mathrm{SK}$ (Lecture 4). Over $\mathbb{Z}$, $\mathrm{ct}^{(i)}(\mathrm{SK}) = \Delta M^{(i)} + v_i + qr_i$. Multiplying,

$$\begin{aligned}\big(\mathrm{ct}^{(1)}\mathrm{ct}^{(2)}\big)(\mathrm{SK}) = \Delta^2 M^{(1)}M^{(2)} + \Delta\big(M^{(1)}v_2 + M^{(2)}v_1\big) + q\big(v_1r_2 + v_2r_1\big) \\ + q\Delta\big(M^{(1)}r_2 + M^{(2)}r_1\big) + v_1v_2 + q^2r_1r_2.\end{aligned} \tag{4}$$

The leading term carries Δ^2, so one factor must come off.

6.2 Scaling by t/q

Dividing by Δ fails on $q^2r_1r_2$: $\Delta \nmid q^2$ in general, and the non-integral residue survives reduction modulo q. Scaling by t/q instead sends $q^2r_1r_2 \mapsto qt\, r_1r_2$, which vanishes modulo q. Since $t/q \approx 1/\Delta$, the leading term still comes out right. Because t/q is not an integer, the result must be rounded, adding an error r_e.

Write $M^{(1)}M^{(2)} = [M^{(1)}M^{(2)}]_t + t\, r_M$ and $v_1v_2 = [v_1v_2]_\Delta + \Delta\, r_v$. Substituting into (4) and scaling by t/q gives, modulo q,

$$\begin{aligned}\tfrac{t}{q}\big(\mathrm{ct}^{(1)}\mathrm{ct}^{(2)}\big)(\mathrm{SK}) = \Delta\big[M^{(1)}M^{(2)}\big]_t + \big(M^{(1)}v_2 + M^{(2)}v_1\big) + t\big(v_1r_2 + v_2r_1\big) + r_v \\ - [q]_t\big(r_M + M^{(1)}r_2 + M^{(2)}r_1\big) + r_e.\end{aligned}$$

This has the standard shape $\Delta M^{(3)} + v_3$ with $M^{(3)} = [M^{(1)}M^{(2)}]_t$. The dominant term of v_3 is $M^{(1)}v_2 + M^{(2)}v_1$: the messages multiply the noises.

Proposition 6 (Multiplicative noise growth)**.** *Up to lower-order terms,*

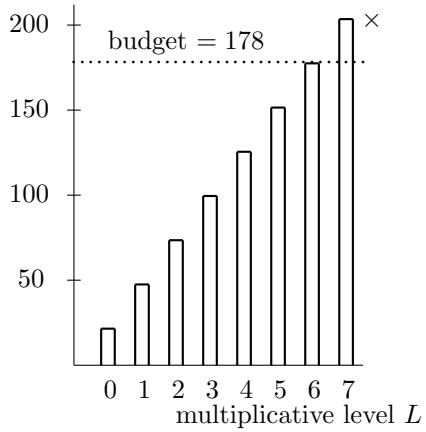
$$\|v_{\mathrm{mult}}\| \;\lesssim\; \|v_i\| \cdot G, \qquad G = 2\,t\,n^2\|\mathrm{SK}\|,$$

where $\|v_i\|$ *is the larger input noise.*

Proof sketch. Since c_0 and c_1 have centered coefficients below $q/2$, $c_0 + c_1\mathrm{SK}$ has coefficients of size at most about $nq\|\mathrm{SK}\|$, so $\|r_i\| \lesssim n\|\mathrm{SK}\|$. The term $t(v_1r_2 + v_2r_1)$ then has norm at most $t \cdot 2 \cdot n\|v_i\| \cdot n\|\mathrm{SK}\| = \|v_i\|\, G$. The other terms are smaller. □

After L sequential multiplications the noise is about $\|v_0\|\, G^L$: exponential in the **depth**, and in nothing else. At $n = 4096$, $t = 2$ and ternary SK, $G = 2 \cdot 2 \cdot 4096^2 = 67\,108\,864 = 2^{26}$, i.e. 26 bits of budget per level.

6.3 A Depth Budget



Bar heights are $\log_2\|v\| = 21.50 + 26L$ after L multiplications at $n = 4096$, $t = 2$, $\beta = 19$, $q = 2^{180}$. The dotted line is the budget $\log_2(q/2t) = 178$, cleared by half a bit at $L = 6$ and overshot by 25.5 bits at $L = 7$.

Example 4 (How many multiplications?)**.** Take $n = 4096$, $t = 2$, $\beta = 19$, ternary SK, $q = 2^{180}$, and the conservative fresh-noise bound.

(a) Noise and growth. $\|v_0\| \le 2n\beta^2 + \beta = 2 \cdot 4096 \cdot 361 + 19 = 2\,957\,331 \approx 2^{21.50}$, and $G = 2 \cdot 2 \cdot 4096^2 \cdot 1 = 2^{26}$.

(b) Depth. The budget is $q/(2t) = 2^{180}/4 = 2^{178}$. We need $21.50 + 26L < 178$, i.e. $L < 156.50/26 \approx 6.02$, so $L = 6$. Check: $21.50 + 6 \cdot 26 = 177.50 < 178$ ✓, while $21.50 + 7 \cdot 26 = 203.50 > 178$ ×.

(c) Depth 12. We need $21.50 + 12 \cdot 26 = 333.50 < \log_2 q - 2$, so $\log_2 q > 335.50$, i.e. $\log_2 q \ge 336$.

(d) Larger t. At $t = 2^{16}$, $G = 2 \cdot 2^{16} \cdot 2^{24} = 2^{41}$ and the budget is $2^{180}/2^{17} = 2^{163}$. Then $21.50 + 41L < 163$ gives $L < 3.45$, so $L = 3$.

6.4 The Procedure

Definition 6 (EvalMult)**.** Tensor, scale by t/q, round and reduce:

$$\mathrm{ct}^* = \left(\left[\left\lfloor \frac{t\, c_0^{(1)} c_0^{(2)}}{q} \right\rceil \right]_q,\ \left[\left\lfloor \frac{t\,(c_0^{(1)} c_1^{(2)} + c_1^{(1)} c_0^{(2)})}{q} \right\rceil \right]_q,\ \left[\left\lfloor \frac{t\, c_1^{(1)} c_1^{(2)}}{q} \right\rceil \right]_q \right).$$

The three components are the coefficients of S^0, S^1, S^2 in $(c_0^{(1)} + c_1^{(1)} S)(c_0^{(2)} + c_1^{(2)} S)$. They are formed over $\mathbb{Z}$ from centered representatives *before* scaling; reducing modulo q first would destroy the information the division by q needs.

The output has two problems: it has **three** ring elements instead of two, and it decrypts only through $c_0^* + c_1^* \mathrm{SK} + c_2^* \mathrm{SK}^2$, which needs SK^2. Both are repaired by relinearization.

Multiplying by a plaintext M' is cheap: $(M'c_0,\ M'c_1)$ needs no tensoring and no rescaling, and the noise grows to at most $n\|M'\|\|v\|$, not by the factor G.

7. Relinearization

7.1 The Evaluation Key

Given $\mathrm{ct}^* = (c_0^*, c_1^*, c_2^*)$ we want $\hat{\mathrm{ct}} = (\hat{c}_0, \hat{c}_1)$ with

$$c_0^* + c_1^*\mathrm{SK} + c_2^*\mathrm{SK}^2 \;\approx\; \hat{c}_0 + \hat{c}_1\mathrm{SK} \pmod q.$$

The evaluator cannot compute $c_2^*\mathrm{SK}^2$, so SK^2 is supplied in masked form.

Definition 7 (Evaluation key)**.** Sample fresh $a \leftarrow R_q$ and $e \leftarrow \mathcal{X}$, and publish

$$\mathrm{EK} = (\mathrm{EK}_0, \mathrm{EK}_1) = \Big(\big[-(a\cdot \mathrm{SK} + e) + \mathrm{SK}^2 \big]_q,\; a \Big).$$

As in (1),

$$\mathrm{EK}_0 + \mathrm{EK}_1 \cdot \mathrm{SK} \;=\; \mathrm{SK}^2 - e. \tag{5}$$

EK is an encryption of SK^2 under SK, so BFV needs a circular-security assumption on top of RLWE.

7.2 Basic Relinearization

Definition 8 (Relinearize, basic form)**.** $\hat{c}_0 = \big[c_0^* + \mathrm{EK}_0 \cdot c_2^*\big]_q$ and $\hat{c}_1 = \big[c_1^* + \mathrm{EK}_1 \cdot c_2^*\big]_q$.

Proposition 7. $\hat{c}_0 + \hat{c}_1\mathrm{SK} = c_0^* + c_1^*\mathrm{SK} + c_2^*\mathrm{SK}^2 - c_2^* \cdot e.$

Proof. $\hat{c}_0 + \hat{c}_1\mathrm{SK} = c_0^* + c_1^*\mathrm{SK} + c_2^*(\mathrm{EK}_0 + \mathrm{EK}_1\mathrm{SK})$; apply (5). □

The error $-c_2^*e$ is a disaster. The key error e is small, but c_2^* is a ciphertext component spread over all of $\mathbb{Z}_q$, so

$$\|c_2^* \cdot e\| \;\le\; n\|c_2^*\|\beta \;\approx\; \tfrac{1}{2}nq\beta,$$

which exceeds q for $n \ge 2$. Basic relinearization destroys the ciphertext.

7.3 Gadget Decomposition

Fix a base w and let $\ell = \lceil \log_w q \rceil$. Decompose c_2^* coefficientwise (Lecture 4):

$$c_2^* = \sum_{i=0}^{\ell-1} w^i d_i, \qquad \|d_i\| < w.$$

Definition 9 (Relinearize with ℓ keys)**.** Publish $\mathrm{EK}^{(i)} = \big([-(a_i\mathrm{SK}+e_i)+w^i\mathrm{SK}^2]_q,\; a_i\big)$ for $0 \le i < \ell$, and set $\hat{c}_j = \big[c_j^* + \sum_i \mathrm{EK}_j^{(i)} d_i\big]_q$ for $j = 0, 1$.

Proposition 8. $\hat{c}_0 + \hat{c}_1\mathrm{SK} = c_0^* + c_1^*\mathrm{SK} + c_2^*\mathrm{SK}^2 - \sum_i d_i e_i$, *and* $\|\sum_i d_i e_i\| \le n\ell w\beta$.

Proof. $\mathrm{EK}_0^{(i)} + \mathrm{EK}_1^{(i)}\mathrm{SK} = w^i\mathrm{SK}^2 - e_i$, so the sum contributes $\sum_i w^i d_i \mathrm{SK}^2 - \sum_i d_i e_i = c_2^*\mathrm{SK}^2 - \sum_i d_i e_i$. Each $\|d_i e_i\| \le nw\beta$, and there are ℓ terms. □

A blow-up of order q is traded for an *additive* term of order w, at ℓ times the key material and ℓ times the work. Since EvalMult has just multiplied the noise by G, the relinearization error is almost never the binding constraint. Choosing w trades key size against noise. Relinearization is an instance of **key switching**, here from SK^2 to SK; the same machinery rotates slots in BGV and CKKS.

8. A Worked Instance

8.1 Setup and Encryption

Example 5 (The instance)**.** Take $n = 4$, $t = 4$, $q = 2^{14} = 16384$, $\beta = 1$, and

$$\mathrm{SK} = 1 - X + X^3, \quad e = 1 - X^2 + X^3, \quad a = 7561 + 3002X + 15115X^2 + 8443X^3.$$

Parameters. $\Delta = 16384/4 = 4096$ and $t \mid q$, so $[q]_t = 0$. The budget is $q/(2t) = 16384/8 = 2048$. Fresh noise is at most $2n\beta + \beta = 9$, and $G = 2 \cdot 4 \cdot 16 \cdot 1 = 128$. One multiplication is guaranteed: $9 \cdot 128 = 1152 < 2048$, and even with the relinearization bound 112 of Section 8.3, $1152 + 112 = 1264 < 2048$. A second is not: $1152 \cdot 128 = 147\,456 > 2048$. Only one level is guaranteed.

Public key. $\mathrm{PK}_0 = [-(a\mathrm{SK} + e)]_q = 3381 + 3290X + 12715X^2 + 15494X^3$ and $\mathrm{PK}_1 = a$; one checks $\mathrm{PK}_0 + \mathrm{PK}_1\mathrm{SK} = -1 + X^2 - X^3 = -e$.

Encode. In base 2, $m_1 = 5$ gives $M^{(1)} = 1 + X^2$ and $m_2 = 2$ gives $M^{(2)} = X$.

Encrypt. With $u^{(1)} = 1 - X^2 + X^3$, $e_1^{(1)} = X - X^2$, $e_2^{(1)} = 1 - X$, and $u^{(2)} = -1 + X + X^2$, $e_1^{(2)} = 1 - X^3$, $e_2^{(2)} = -X + X^2 + X^3$, (2) gives (coefficients in $[0, q)$)

$c_0^{(1)}$	$518 + 6070X + 14319X^2 + 15585X^3$
$c_1^{(1)}$	$3291 + 12713X + 15495X^2 + 13002X^3$
$c_0^{(2)}$	$1179 + 5077X + 10340X^2 + 510X^3$
$c_1^{(2)}$	$1649 + 12499X + 11833X^2 + 9675X^3$

Decrypt. Computing $[c_0 + c_1\mathrm{SK}]_q - \Delta M$, and independently $-eu + e_1 + e_2\mathrm{SK}$, gives the same noise in each case:

$$v_1 = 2 - 3X + 3X^2 - X^3, \quad \|v_1\| = 3, \qquad v_2 = 4 - 2X - X^2 + X^3, \quad \|v_2\| = 4.$$

Both are below 9, so (3) returns $M^{(1)}$ and $M^{(2)}$, which decode to 5 and 2.

8.2 One Addition, One Multiplication

Example 6 (Add and multiply)**.** EvalAdd. The noise is $v_1 + v_2 = 6 - 5X + 2X^2$, with norm $6 \le \|v_1\| + \|v_2\| = 7$. Decryption returns $1 + X + X^2$, which decodes to $1 + 2 + 4 = 7 = 5 + 2$.

EvalMult. The three components are

$$\begin{aligned} c_0^* &= 13721 + 1467X + 6265X^2 + 4702X^3, \\ c_1^* &= 11383 + 12209X + 15054X^2 + 14179X^3, \\ c_2^* &= 7501 + 6571X + 10312X^2 + 14554X^3. \end{aligned}$$

Evaluating $c_0^* + c_1^*\mathrm{SK} + c_2^*\mathrm{SK}^2$ gives $\Delta M^{(3)} + v_3$ with

$$M^{(3)} = X + X^3, \qquad v_3 = 7 - 2X + X^2 + 2X^3, \qquad \|v_3\| = 7.$$

$M^{(3)}$ decodes to $2 + 8 = 10 = 5 \times 2$. The noise grew from 4 to 7, against the worst case $\|v_2\| \cdot G = 4 \cdot 128 = 512$: worst-case bounds are very pessimistic, which is the price of a guarantee.

8.3 Relinearization

Example 7 (Relinearize, both ways)**.** Take $a' = 11923{+}604X{+}9981X^2{+}14002X^3$ and $e' = X{-}X^2$.

Key. $\mathrm{EK}_0 = [-(a'\mathrm{SK} + e') + \mathrm{SK}^2]_q = 7450 + 4913X + 4626X^2 + 442X^3$, $\mathrm{EK}_1 = a'$, and $\mathrm{EK}_0 + \mathrm{EK}_1\mathrm{SK} - \mathrm{SK}^2 = -X + X^2 = -e'$, as (5) states.

Basic. By Proposition 7 the new noise is $[v_3 - c_2^* e']_q = 4249 - 5673X + 931X^2 - 3739X^3$, so $\|v\| = 5673 > 2048$. Decryption returns 1 instead of $X + X^3$: the message is destroyed. The bound predicted it: in centered form $c_2^* = 7501{+}6571X{-}6072X^2{-}1830X^3$, so $n\|c_2^*\|\beta = 4{\cdot}7501 = 30\,004$.

Gadget, $w = 4$. Here $\ell = \lceil \log_4 16384 \rceil = 7$, so seven keys. The base-4 digits of c_2^* (coefficients in $[0, q)$) are

$$\begin{aligned} &d_0 = 1 + 3X + 2X^3, && d_1 = 3 + 2X + 2X^2 + 2X^3, && d_2 = 2X + X^3, \\ &d_3 = 1 + 2X + X^2 + 3X^3, && d_4 = 1 + X, && d_5 = 3 + 2X + 2X^2 + 2X^3, \\ &d_6 = 1 + X + 2X^2 + 3X^3, \end{aligned}$$

and $\sum_i 4^i d_i = c_2^*$. For the toy, all seven keys reuse a' and e' (insecure, but the noise depends only on the e_i). Then $\sum_i d_i e_i = \big(\sum_i d_i\big)e'$ with $\sum_i d_i = 10 + 13X + 7X^2 + 13X^3$, so $\sum_i d_i e' = -6 + 23X + 3X^2 - 6X^3$ and the new noise is

$$v_3 - \sum_i d_i e' = 13 - 25X - 2X^2 + 8X^3, \qquad \|v\| = 25,$$

against $\|v_3\| + n\ell w\beta = 7 + 4 \cdot 7 \cdot 4 = 119$. Decryption returns $X + X^3$. The relinearized ciphertext is

$$\hat{c}_0 = 8279 + 2761X + 4059X^2 + 9120X^3, \qquad \hat{c}_1 = 1940 + 11017X + 7767X^2 + 811X^3.$$

8.4 Where It Breaks

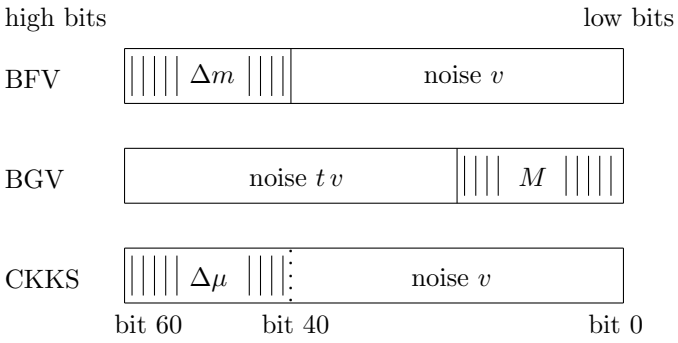
Example 8 (A second multiplication)**.** Multiply $\hat{\mathrm{ct}}$ by $\mathrm{ct}^{(2)}$ again, to compute $5 \times 2 \times 2 = 20$. Decrypting the three-element result gives noise $-8 - 5X - 19X^3$, of norm 19, far inside the budget 2048, and

$$M^{(4)} = \big[(X + X^3) \cdot X\big]_t = X^2 + X^4 = -1 + X^2 = 3 + X^2 \text{ in } R_4.$$

So $M^{(4)} = M^{(3)}M^{(2)}$ in R_t: BFV worked. But decoding at $X = 2$ gives $3 + 4 = 7$, not 20. The encoding failed: $20 = 10100_2$ needs degree 4, which $n = 4$ does not have (condition (b)), and the wrap $X^4 \equiv -1$ put $-1 \equiv 3 \pmod 4$ in the constant term.

A computation can therefore fail in two independent ways — through the noise budget, or through the plaintext geometry — and neither announces itself. Both must be checked when the parameters are chosen.

9. Where the Message Sits



The word is $\log_2 q = 60$ bits wide with $t = 2^{20}$, $\Delta = 2^{40}$; hatching marks the message. BFV puts it in the top twenty bits, BGV in the bottom twenty, CKKS in the top with a dotted boundary, there being no separation at all.

	BFV	**BGV**	**CKKS**
Decryption	$c_0 + c_1 SK$ $= \Delta m + v$	$c_0 + c_1 SK$ $= M + tv$	$c_0 + c_1 SK$ $= \Delta\mu + v$
Recover by	scaling by t/q, rounding, then mod t	reducing mod t; no rounding	dividing by Δ and accepting the error
Correct when	$\|v\| < q/(2t)$	$\|v\| < q_\ell/(2t)$	never exactly; precision about $\log_2 \Delta - \log_2 \|v\|$, in $\mathbb{C}^{n/2}$

9.1 Reading the Table

BFV keeps q fixed and tracks the invariant noise tv/q, scaling down by t/q after each multiplication; at $q = 2^{30}$, $t = 2^{10}$ the message holds the top ten bits under $\Delta = 2^{20}$ and the budget is 2^{19}. BGV's error always appears as $t \cdot e$, so it never touches the message's residue modulo t and decryption is a reduction, at the cost of noise growing toward $q_\ell/2$ and shrunk by modulus switching from q_ℓ to $q_{\ell-1} = q_\ell/p_\ell$: at $q_\ell = 2^{60}$, $t = 2^{20}$ and $\|v\| = 2^{35}$ against a budget 2^{39}, switching by $p_\ell = 2^{20}$ leaves 2^{15} against 2^{19}, the same four-bit gap, and each $p_i \equiv 1 \pmod t$ so the message survives. CKKS does not separate message from noise: the plaintext is $\Delta\mu$ under the canonical embedding of Lecture 4 and v costs $\log_2\|v\|$ bits of precision, so $\Delta = 2^{40}$ with $\|v\| = 2^{12}$ keeps 28 bits, and RESCALE by $p_\ell = 2^{40}$ undoes a multiplication's scale squaring and divides the noise in one step — so its decryption returns μ plus the error, carrying information about SK.

Summary

1. Plaintexts lie in R_t and ciphertexts in $R_q \times R_q$; $\mathrm{PK}_0 + \mathrm{PK}_1\mathrm{SK} = -e$. Base-$k$ encoding needs coefficients below t and degree below n.
2. $c_0 + c_1\mathrm{SK} = \Delta M + v$ with $v = -eu + e_1 + e_2\mathrm{SK}$; fresh noise is at most $2n\beta + \beta$ (conservatively $2n\beta^2 + \beta$).
3. Decryption is correct while $\|v\| < q/2t$, and failure is silent.

4. EvalAdd adds noises. EvalMult scales the tensor product by t/q and multiplies the noise by $G = 2tn^2\|\mathrm{SK}\|$, so depth is the scarce resource.
5. Relinearize absorbs $c_2^*\mathrm{SK}^2$ using EK, at error $-c_2^*e$; gadget decomposition cuts it from about $nq\beta$ to $n\ell w\beta$.

Exercises

1. Compute $(1 + 2X + X^2 - X^3)(1 + X + X^2 + X^3)$ in R_5 with $n = 4$ (Example 1), using $X^4 \equiv -1$.

2. Encode $m = 6$ in base 2 at $n = 4$, square the polynomial in R_4, and decode. Which compatibility condition fails, and how does the answer change at $n = 8$?

3. Verify (1) for the keys of Example 5. Then encrypt $M = 1 + X$ with $u = 1$, $e_1 = e_2 = 0$, decrypt, and compute v both from $[c_0 + c_1\mathrm{SK}]_q - \Delta M$ and from $-eu + e_1 + e_2\mathrm{SK}$.

4. Take $q = 100$, $t = 7$, so $\Delta = 14$ and $[q]_t = 2$. Show that $\Delta \nmid q^2$, and divide the term $q^2 r_1 r_2$ of (4) by Δ and by q/t in turn. Which result is removed by reduction modulo q?

5. For $(n, t) = (4096, 2)$, $(4096, 256)$ and $(8192, 2)$ with ternary SK, compute G in bits and the depth supported at $q = 2^{180}$, $\beta = 19$, using the conservative fresh-noise bound. Which of n, t, q buys depth most cheaply?

6. In Example 5, how many fresh ciphertexts can be added before the worst-case bound $L(2n\beta + \beta)$ reaches $q/2t$? Compare with the guaranteed multiplicative depth.

7. For $q = 16384$, $n = 4$, $\beta = 1$, tabulate ℓ and $n\ell w\beta$ for $w = 2, 4, 16, 128$. What is paid in key material as w decreases? Then find the smallest power-of-two n and the smallest t for which $5 \times 2 \times 2$ decodes correctly under base-2 encoding.